

CRAFT — Pitch Package

Audited retrieval-augmented AI assistant for CSA mission engineering and finance (bilingual EN/FR, cited answers, HITL-gated).

1. Containers (L2)

Figure 1 · L2 — Containers, Actors, External Systems

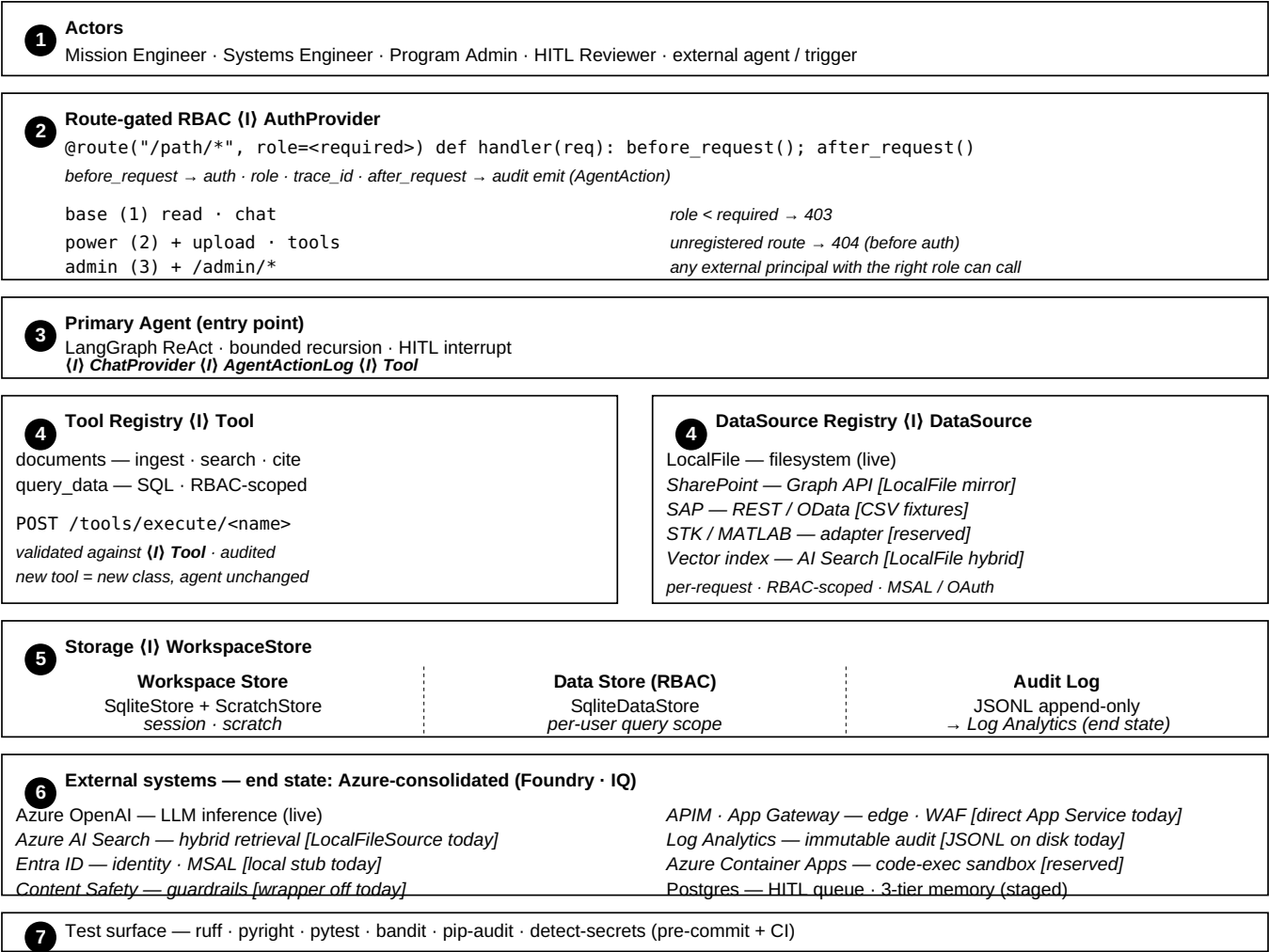


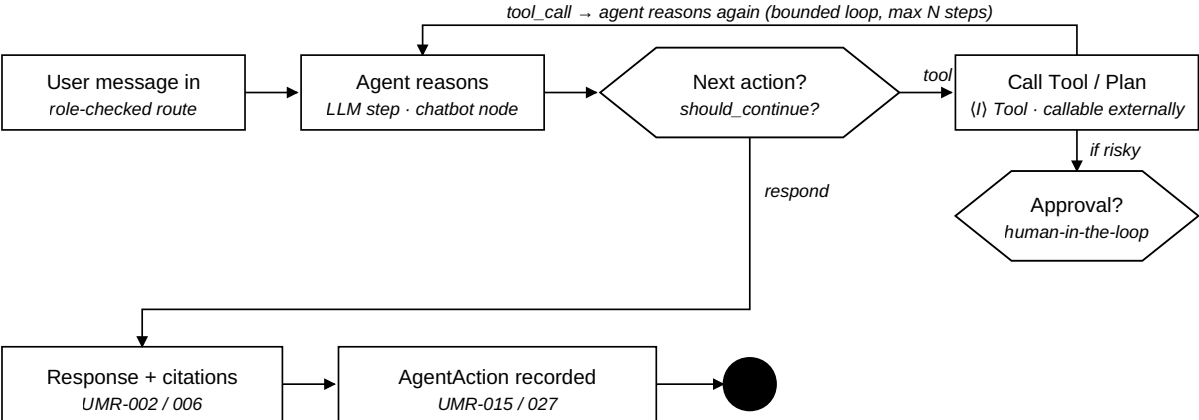
Figure 1 · Actors, routes, containers, data, external systems, audit perimeter.

Flask + Dash. Routes are role-checked; every step emits AGENT_ACTION (UMR-015/027). LangGraph ReAct dispatches to the Tool and DataSource registries; responses carry citations (UMR-002/006) and confidence (UMR-007/010).

2. Internals (L3)

Figure 2 · L3 — Internal Clusters

A · Primary Agent — what happens inside one turn



B · Tool Registry (I) Tool

examples: documents · query_data

POST /tools/execute/<name>

body: { "args": {...}, "trace_id": ... }

validated against (I) Tool · audited

a Tool may itself be a planning sub-agent

C · DataSource Registry (I) DataSource

LocalFile (live) · others pending (see Figure 1)

concrete backend is swappable per-request

per-request · RBAC-scoped · MSAL / OAuth

response carries doc / page / section + confidence

end state: Azure AI Foundry index / Azure AI Search

route → before_request (auth · role · trace_id) → handler → after_request (audit emit) → response

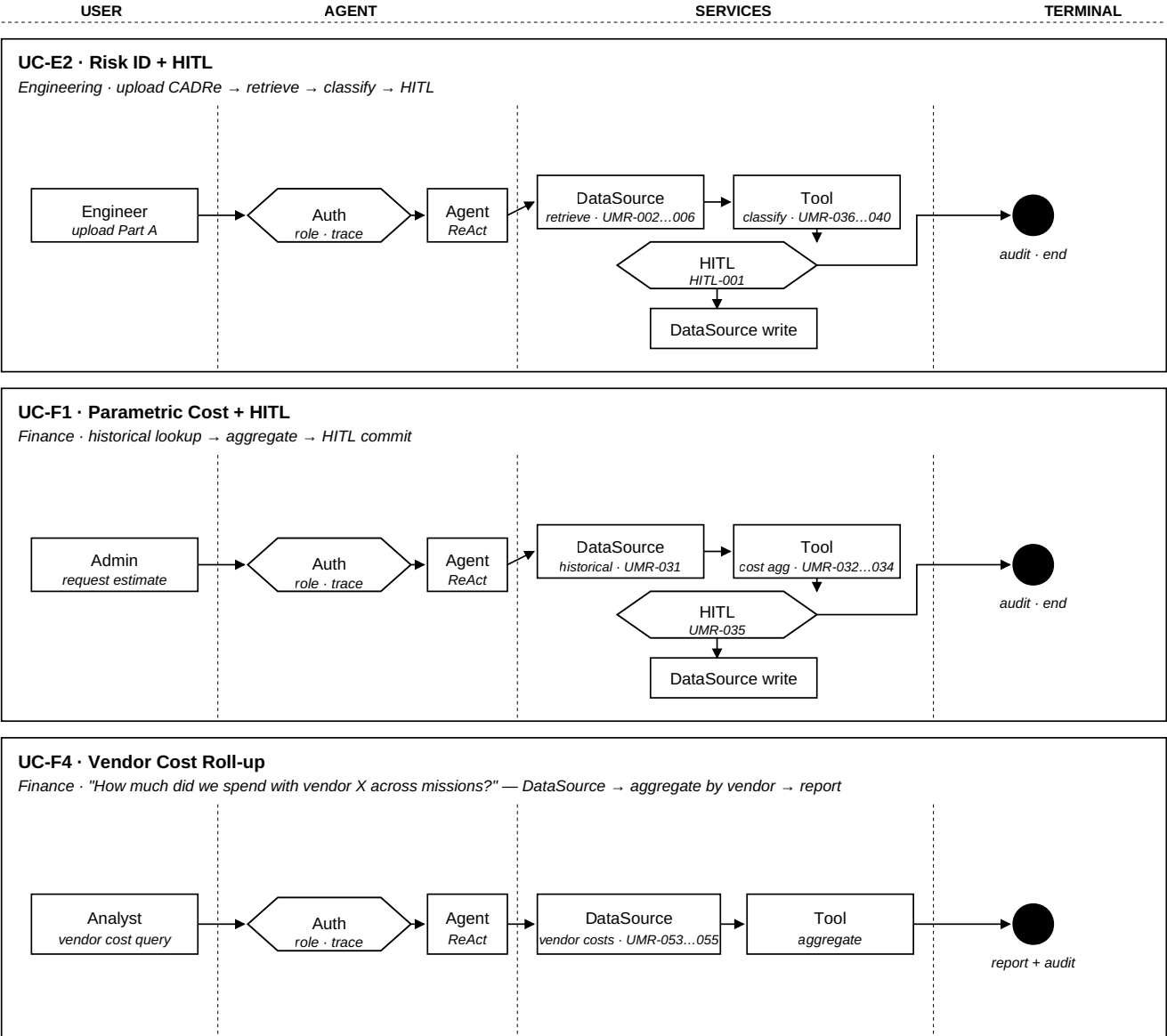
base(1) < power(2) < admin(3) · role < required → 403 · unregistered route → 404 before auth

Figure 2 · Agent loop with HITL branch, Tool registry, DataSource registry, auth cluster.

HITL (UMR-021–026/HITL-001–007) is a graph interrupt off tool_node, not middleware: the queue, timeout, reviewer UI, and decision log sit inside the same auth and audit perimeter as the agent. Self-correction / grounding loops (UMR-009/010) attach as graph nodes between retrieve and respond.

3. Use Cases

Figure 3 · Use-Case Lanes — E2 · F1 · F4



Service blocks are the **(I) Tool** and **(I) DataSource** abstractions — the concrete provider sits behind them and is swappable.
Role-check · AgentAction audit · HITL interrupt stay fixed across use cases, so new flows land safely.

Figure 3 · UC-E2 risk + HITL · UC-F1 parametric cost + HITL · UC-F4 vendor cost roll-up (SAP).

4. Design Decisions

COMPONENT	END-STATE CHOICE	CURRENT ALTERNATIVE	WHY
HTTP surface	Flask + Dash reviewer UI	— (live)	Routes are the integration contract — any token-bearing caller (user, cron, agent) can hit CRAFT; Dash shares the auth and audit perimeter.
Agent runtime	LangGraph ReAct	— (live)	First-class interrupt for HITL, bounded recursion, graph-level test anchors; maintained by LangChain Inc.
LLM inference	Azure AI Foundry (Azure IQ) — model routing + rate limits + Content Safety	Azure OpenAI direct (Canadian PBMM region)	Foundry collapses model, rate-limit and guardrail procurement into one line while PBMM residency stays.
Retrieval	Azure AI Foundry index / Azure AI Search hybrid (vector + BM25)	LocalFileSource via {I} DataSource with filename / path search	Same {I} DataSource call site; cutover is config. Satisfies UMR-004/005/006.
Guardrails + sandbox	Content Safety + Azure Container Apps	Guardrail hook disabled; CodeExecTool slot reserved	End-state wires into the tool wrapper CRAFT already runs. UMR-020 / UMR-093.
Identity	Entra ID + MSAL	Local AuthProvider stub	Same token flow for humans and external agents; CSA-standard.
Audit sink	Azure Log Analytics (immutable retention)	Hash-chained JSONL on disk	KQL at end state; grep today; UMR-027 / ASG-003.
HITL queue + memory	Postgres (approval queue + 3-tier memory)	SQLite scratch; approval table staged	Durable across restarts; UMR-025 TTL, UMR-014 memory.
Session + per-user data	Managed DB behind {I} WorkspaceStore	SQLite file-local	Swap target already named behind the protocol.
Edge	APIM + App Gateway (WAF)	Direct App Service	WAF + rate limit at perimeter; UMR-030.
Deployment	App Service PBMM + SSC LaunchPad HA	Single-region App Service	HA + landing zone come with LaunchPad; UMR-057.
Protocol contracts	{I} Tool, {I} DataSource, {I} ChatProvider, {I} WorkspaceStore, {I} AgentActionLog	— (live)	New capability or backend = one class satisfying a protocol; agent unchanged. Pyright-checked.
Invariants	pytest + pyright + ruff + bandit + pip-audit + detect-secrets	— (live)	Route × role matrix, protocol contracts, audit-envelope present. Industry-standard, maintained.

5. Solving the Problems — Today and at End State

The system is already running against stubs or local alternatives wired through the same registries that will carry the Azure services at end state. Cutover is a config change per row, not a rewrite.

PROBLEM (REQUIREMENT)	HOW THE ARCHITECTURE SOLVES IT TODAY	HOW IT SOLVES IT AT END STATE (AZURE-CONSOLIDATED)
Cited retrieval (UMR-002/005/006, ARR-002/003)	DocumentSearchTool + TextAnalysisTool over LocalFileSource; retrieval response schema carries document / page / section provenance.	Replace LocalFileSource with Azure AI Search backend (hybrid semantic + BM25); same DataSource call site, same citation schema.
Grounding + confidence (UMR-007/008/010)	Grounding node sits between retrieve and respond; emits confidence into AGENT_ACTION; escalates below 0.6.	Foundry evaluation + RAGAS-style scoring plugs into the same node; threshold remains code-side.
HITL approvals (UMR-021–026, HITL-001–007)	LangGraph interrupt → Postgres approval queue → Dash reviewer UI, same auth as routes; decisions logged immutably with reasoning chain, confidence, sources.	Content Safety + Foundry rate limits layer in front; approval queue and UI unchanged.
RBAC + data scope (UMR-012/017/058, AAR-002/007)	Route-gated _ROUTE_RULES + _ROLE_LEVEL; SqliteDataStore scoped by QueryTool.set_user; per-tool RBAC column staged.	Entra ID issues tokens; same role check in before_request; classification tags enforced at the data store.
Immutable audit (UMR-015/027/045, ASG-003)	AgentActionLog writes append-only JSONL with hash chain; daily rotation; covers every tool call and LLM turn.	Forward the same envelope to Log Analytics (immutability policy); KQL replaces grep.
Guardrails + sandbox (UMR-020/029/030/093, ASG-001/004/011)	Guardrail hook point in tool wrapper; code-exec slot reserved; rate limit at APIM-like proxy.	Content Safety on the hook; Azure Container Apps for code-exec; Foundry rate controls.
External data (UMR-051–054)	LocalFileSource reads mirrors / CSV fixtures of the CSA corpus and SAP line items; UC-E2 / F1 / F4 lanes run end-to-end.	SharePoint, SAP, STK, MATLAB each land as a DataSource class with MSAL / OAuth.
Multi-agent + memory (UMR-011/014/018/095, AAR-001/004)	Primary agent + tool registry + session workspace; agent_id in AGENT_ACTION already carries sub-task identity.	Add orchestrator + specialist nodes; Postgres holds the three memory tiers; per-session agent factory.
Bilingual reports (UMR-001/046/048)	Agent accepts EN/FR free text; report generator attaches citations and runs through HITL.	Azure Translator + Foundry NLG for narrative sections; same template + evidence pipeline.
Interoperability (UMR-055)	REST surface is the contract today; schema documented; external agents can trigger CRAFT with a token.	Same routes behind APIM + App Gateway (WAF, rate limit); GraphQL layer optional on top.
SLA + scale (UMR-056/057/059)	Single-instance App Service with health checks; load-test harness in place.	HA App Service + Azure Front Door; auto-scale; SSC LaunchPad for the landing zone.

6. Requirements Traceability

CAPABILITY	REQUIREMENTS	DESIGN
Primary agent	UMR-001/011/013/016/018/019/094/095, AAR-001/003/006, ASG-005	LangGraph ReAct · ChatProvider · bounded recursion
Retrieval + RAG	UMR-002/003/004/005/006/009/010/060/067, ARR-002-007	Hybrid search tool · DataSource · citation + grounding schema
Audit + traceability	UMR-015/027/045/061/062/093, AAR-005, ASG-003	AgentAction envelope · AgentActionLog · hash-chained JSONL → Log Analytics
HITL approvals	UMR-021-026/035/039/044/049, HITL-001-007, ASG-002	LangGraph interrupt · Postgres queue · Dash reviewer UI
RBAC + data scope	UMR-012/017/058/065/066, AAR-002/007	_ROUTE_RULES + _ROLE_LEVEL · SqliteDataSource · QueryTool.set_user
Session + workspace	UMR-014/092/095	WorkspaceStore → SqliteStore + SessionScratchStore; per-session agent factory
Guardrails + containment	UMR-020/029/030, ASG-001/004/011	Content Safety hook · APIM rate limit · Container Apps sandbox
Connectors + external	UMR-051-055	DataSource + MSAL · Flask REST · SAP / STK / MATLAB classes
Domain capabilities	UMR-007/008/031-050/075, ARR-001	New Tool classes over existing registries
Deployment + SLA	UMR-028/056/057/059/064/069/070/074/085, ASG-006	Azure App Service PBMM · APIM + WAF · SSC LaunchPad HA · git config

7. Blockages (End State / Current Alternative)

Each row is an Azure-consolidated end-state component wired through the registry the current alternative already runs against.

END-STATE COMPONENT	CURRENT ALTERNATIVE
Azure AI Search (hybrid retrieval)	LocalFileSource + file-name/path search
Azure Log Analytics + immutability	Hash-chained append-only JSONL on disk
Azure Content Safety	Guardrail hook off (config flag)
Azure Container Apps sandbox	CodeExecTool registration slot reserved
Entra ID (app registration + MSAL)	Local AuthProvider stub, mocked tokens
SharePoint DataSource	LocalFileSource reads a corpus mirror
SAP DataSource	CSV fixtures via LocalFileSource (UC-F4 runs)
STK / MATLAB adapters	Tool slot reserved
Azure AI Foundry (model routing / rate limits)	Azure OpenAI direct (Canadian region)
SSC LaunchPad HA	Single-region App Service
CSA risk taxonomy	Stub taxonomy drives the classifier for tests
Historical mission DB	CSV fixtures through LocalFileSource (UC-F1 runs)

8. System-Health Tooling

Runs on every commit and in CI; these are invariants, not blockages.

TOOL	ROLE
ruff	Lint (flake8 + isort + pyupgrade)
pyright (strict)	Type checking · protocol conformance
pytest + coverage	Route × role matrix · protocol contracts · audit-envelope invariants (156 tests, 91% cov)
bandit	Security scan
pip-audit	Dependency CVE audit
detect-secrets	No credentials in git